

Reinforcement Learning Laboratory

Lab Assignment 4: MATLAB Reinforcement Learning Onramp

Course Outcomes: CO1, CO2, CO4

Title

MATLAB Reinforcement Learning Onramp: Agent Training and Learning Curve Analysis

Objective

The objective of this assignment is to introduce students to the MATLAB Reinforcement Learning environment through the Reinforcement Learning Onramp course. Students will complete the guided training modules, train an RL agent, observe the agent's learning behaviour, and analyze the resulting learning curves.

Reference

MATLAB Reinforcement Learning Onramp. Students should access the course through their MATLAB/MathWorks account.

Instructions

1. Log in to your MathWorks/MATLAB account using your institutional or registered account.
2. Open the Reinforcement Learning Onramp course.
3. Complete all modules/sections provided in the Onramp course.
4. Follow the examples and exercises provided in the course and do not skip the programming exercises.
5. Execute the MATLAB code provided in the course and understand the purpose of each major step.
6. Complete the exercises and assessments associated with the course.
7. Train the Reinforcement Learning agent provided in the Onramp exercise.
8. Observe how the agent's performance changes as training progresses.
9. capture the learning curve/training progress plot generated during agent training.
10. Analyze the learning curve by identifying initial learning behaviour, improvement, reward variation, approximate convergence, and final performance.
11. Complete the Onramp course and obtain the course completion certificate.
12. Take screenshots of course completion, certificate, RL agent training, and learning curve.
13. Submit the required files

Task 1: Complete MATLAB Reinforcement Learning Onramp

- Complete all learning modules.
- Complete the interactive exercises.
- Execute the MATLAB examples.
- Complete the assessments/quizzes provided in the course.

- Obtain the course completion certificate.

Submission: Submit a screenshot of the completed course/certificate along with the certificate file, if available.

Task 2: Train an RL Agent in MATLAB

14. Create or initialize the given RL environment.
15. Create the RL agent.
16. Configure the training options.
17. Train the agent.
18. Observe the training progress.
19. Record the final training performance.

Observation Table

Parameter	Observation
RL Environment	
Agent Used	
Number of Training Episodes	
Initial Performance	
Final Performance	
Training Time	
Training Stopped/Converged At	

Questions

20. What type of RL environment is used in the Onramp exercise?
21. What type of agent is trained?
22. What is the purpose of training the agent?
23. What performance measure is used during training?
24. How does the agent's performance change as training progresses?

Task 3: Analyze the Learning Curve

Analyze the learning curve generated during agent training. Identify the initial reward/performance, maximum reward, average reward trend, improvement, stabilization, and approximate convergence.

Training Stage	Episode Range	Reward/Performance	Observation
Initial			
Early Training			
Middle Training			
Late Training			
Final			

Analysis Questions

25. What does the learning curve represent?
26. Why is the reward generally low during the initial training episodes?
27. What indicates that the agent is learning?

28. Does the reward increase consistently throughout training? Explain.
29. How can you identify convergence from the learning curve?
30. What could cause fluctuations in the learning curve?
31. What happens if the agent is trained for a larger number of episodes?
32. Why is it important to analyze the learning curve rather than considering only the final reward?

Certificate Submission

- MATLAB Reinforcement Learning Onramp Certificate
- Screenshot of the completed Onramp course
- Screenshot of the trained RL agent/training progress
- Screenshot of the learning curve
- Completed observation tables
- Answers to the analysis questions

Microsoft Teams Submission Instructions

33. Open the Microsoft Teams course/team for Reinforcement Learning Laboratory.
34. Navigate to the assignment titled: Lab Assignment 4 – MATLAB Reinforcement Learning Onramp.
35. Upload all required files.
36. Ensure that your name and roll number are clearly visible on the submitted documents.
37. Click Turn in/Submit after uploading all required files.
38. Verify that the submission status shows Submitted/Turned in.

Suggested file naming format:

File name: RollNo_Name_CSE/DS/Phd

The file should include the followings

- RL_Onramp_Certificate
- Training_Result
- Learning_Curve
- R_Assignment4

Important

- Only students who have completed the Onramp course should submit the certificate.
- Do not submit another student's certificate.
- The certificate should clearly display the student's name and course completion.
- Training results and learning-curve screenshots should correspond to the student's own MATLAB execution.

- Students should retain the MATLAB code and results for verification during the laboratory session.

Expected Learning Outcome

Upon successful completion of this assignment, students will be able to navigate and use the MATLAB Reinforcement Learning environment, understand the workflow involved in training an RL agent, train an RL agent using MATLAB, interpret training progress and learning curves, identify improvement and convergence, and relate practical RL implementation to concepts learned in the course.